Question 6: Implement a basic Jenkins CI/CD pipeline that pulls source code from the GitHub repository and triggers an automated build.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-1-

1. Credential setup

stored credential

```
Manage Jenkins -> Credentials -> System -> Global -> Add Credentials
Kind:          Username with password
Username:      meenakshi16rp
Password:      ghp_**************** (GitHub PAT, repo scope)
ID:            github-creds
```

2. Job configuration

job configuration

```
New Item -> ensp461-lab1-pipeline -> Pipeline
Definition:     Pipeline script from SCM
SCM:            Git
Repository URL: https://github.com/meenakshi16rp/Devops-lab-1-.git
Credentials:    github-creds
Branch:         */main
Script path:    Jenkinsfile
Build trigger:  Poll SCM - H/5 * * * *
```

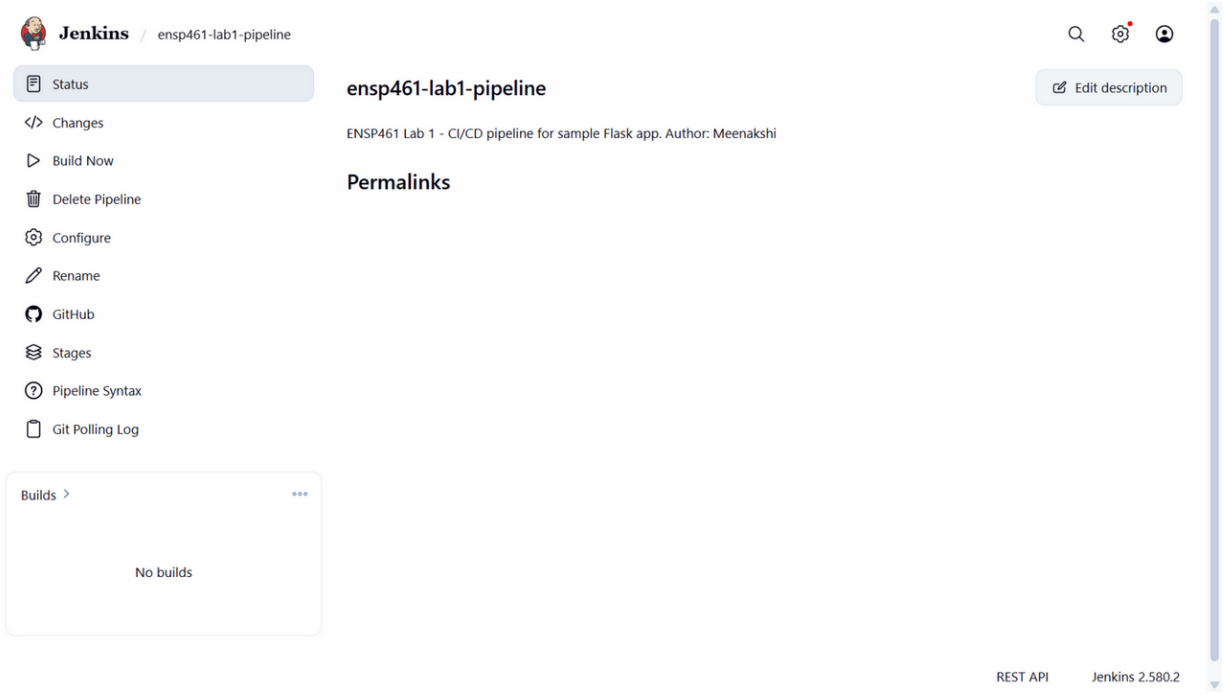


Figure: Pipeline job page after configuration (GitHub project link, stages, polling log)

3. First automated build (and a real defect)

Build #1 pulled the source from GitHub and passed Checkout, Build and Test, but failed in the Deploy stage with exit code 7 (curl connection refused). The console log and the deploy.log identified the cause: the deploy script started the app with the system Python, which has no Flask module.

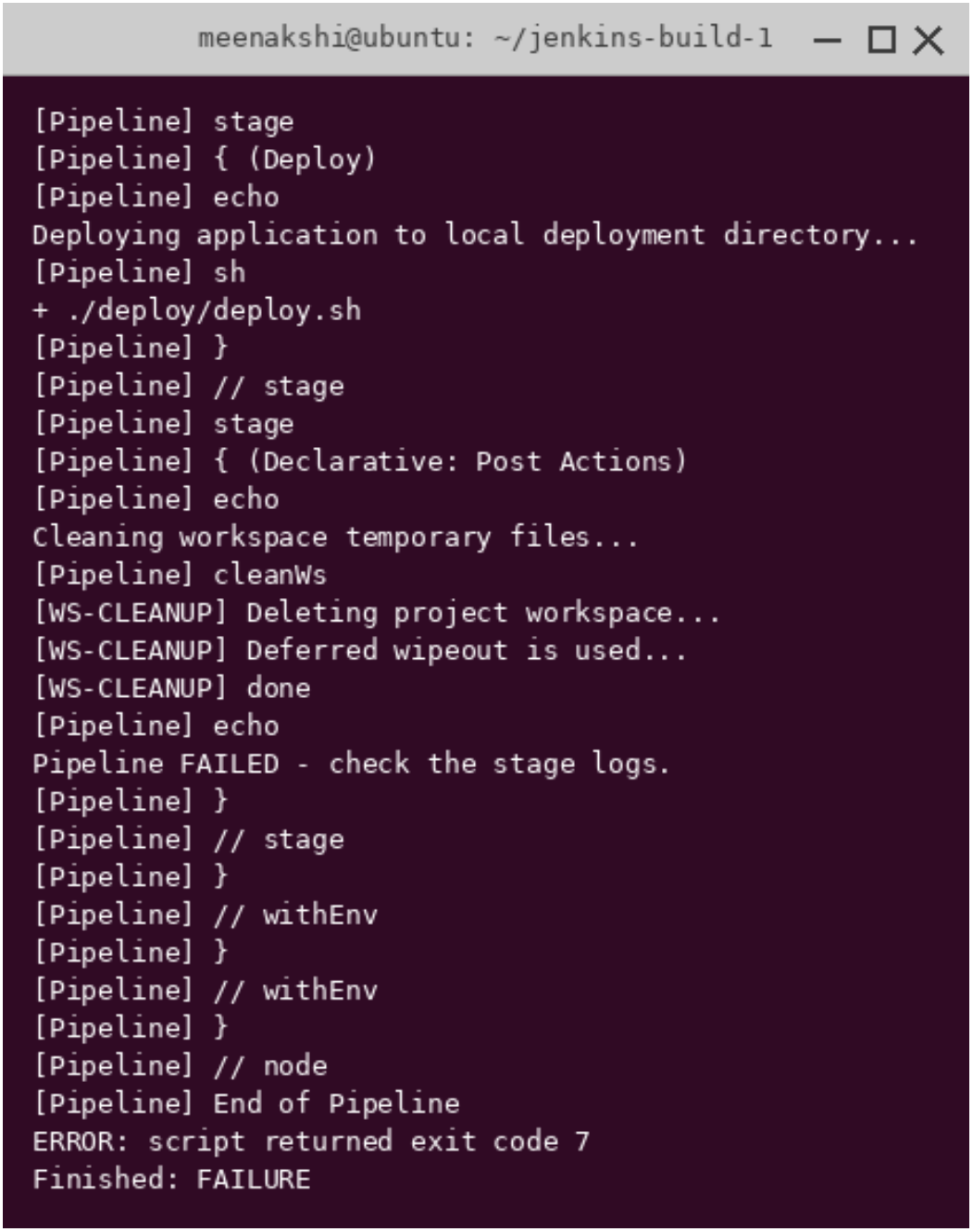


Figure: Build #1 console output (tail) - FAILURE

- Root cause: ModuleNotFoundError: No module named 'flask' in the deployment runtime.
- Fix: deploy.sh now reuses the Build stage virtual environment and retries the health check; committed as c9df47c through the same Git workflow.

4. Successful build

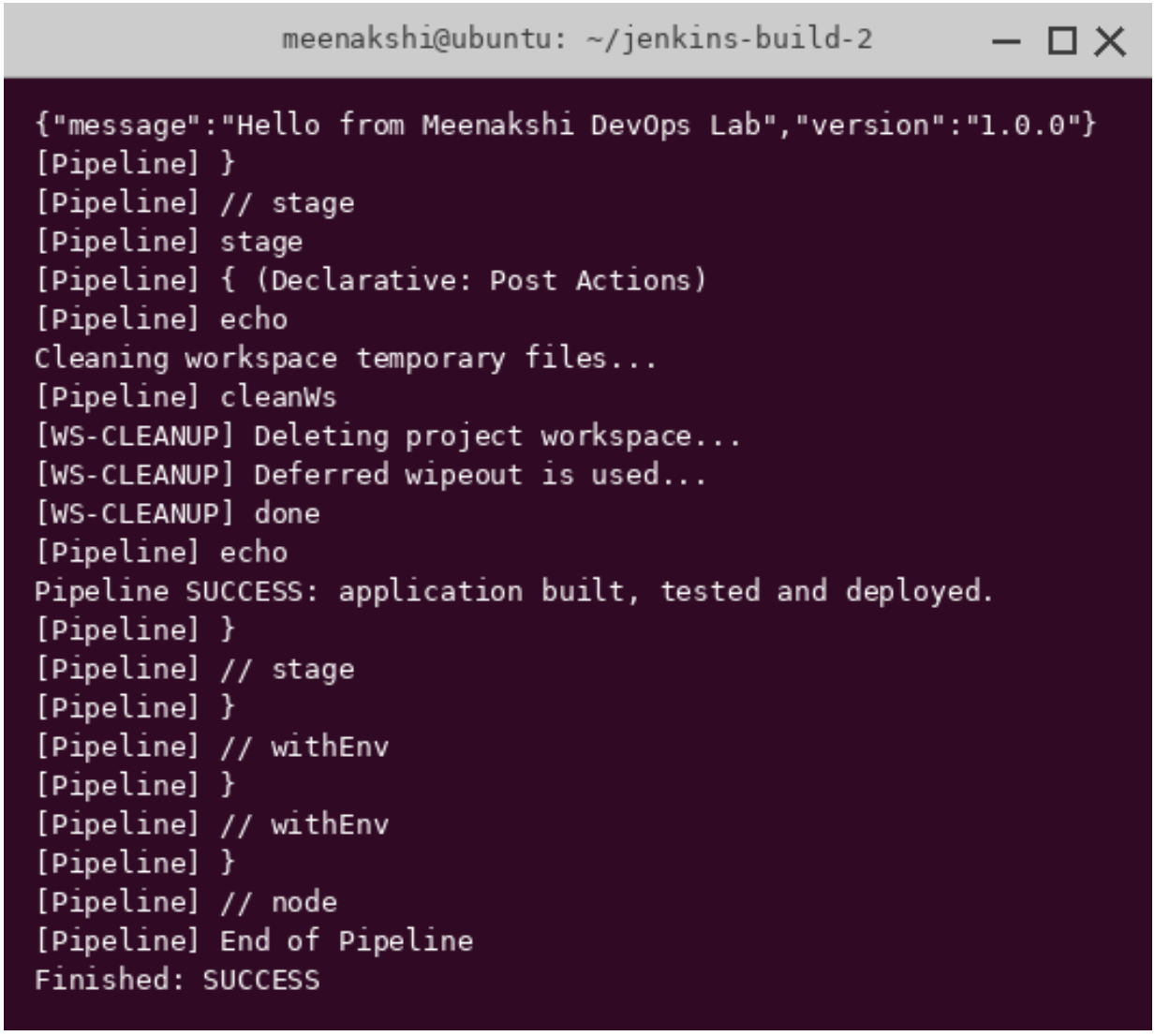


Figure: Build #2 console output (excerpt) - SUCCESS

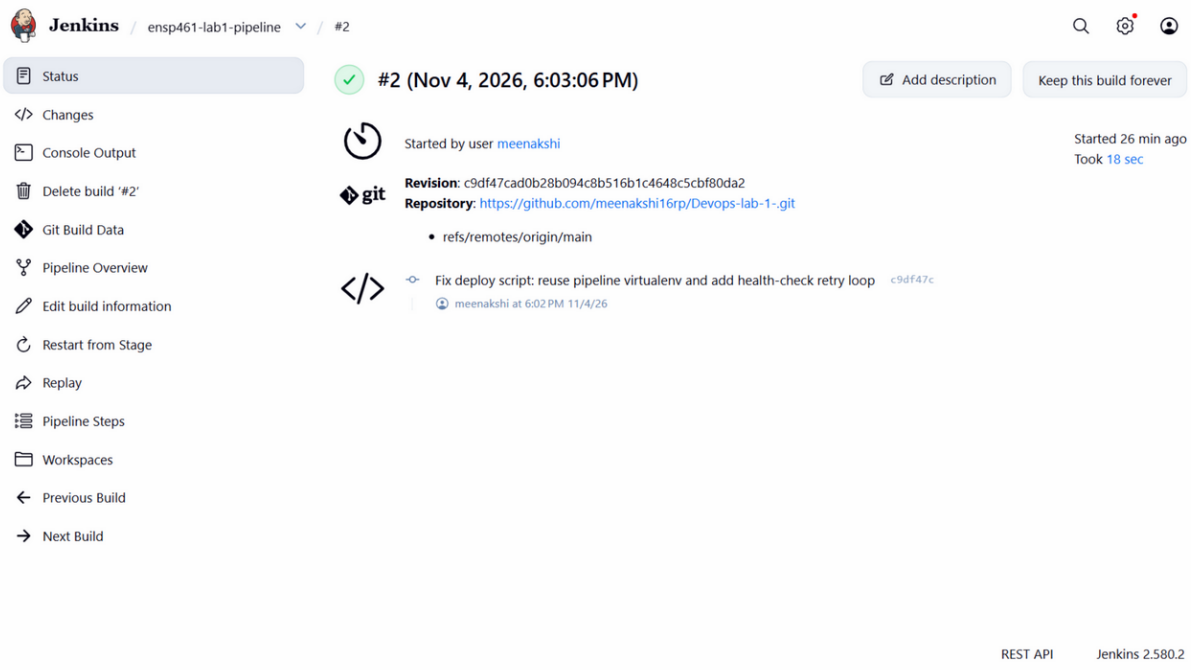


Figure: Build #2 result page: SUCCESS, started by user meenakshi, commit c9df47c

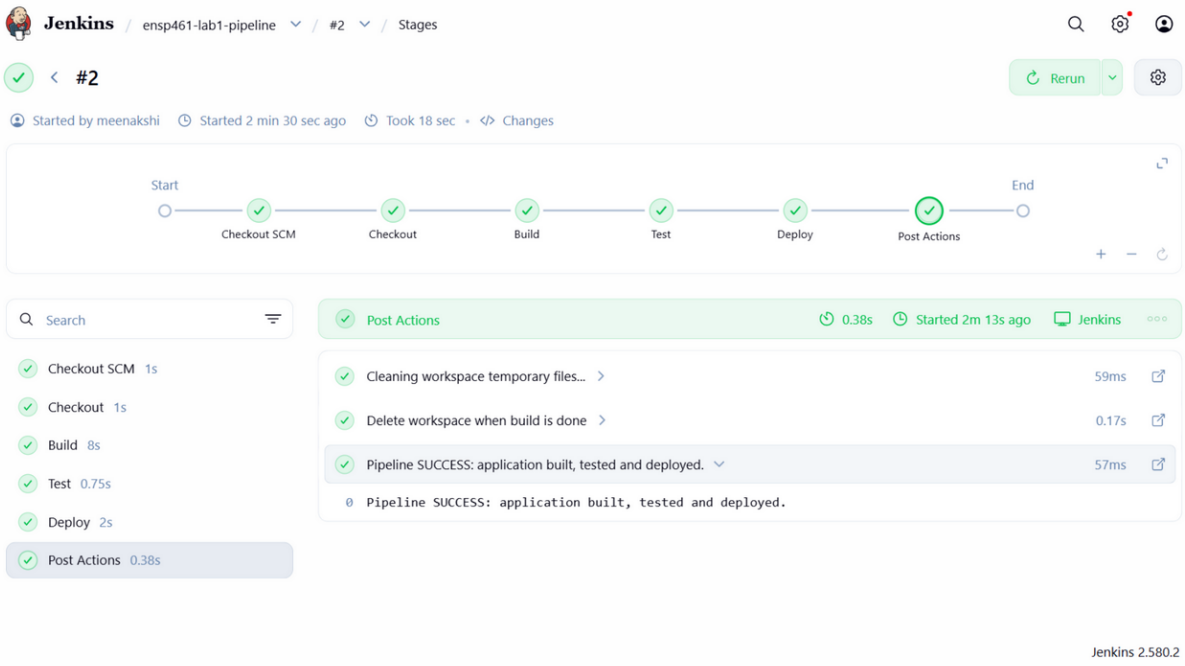


Figure: All pipeline stages green for build #2

5. Observations

- Pipeline-as-code means the job configuration in Jenkins stays minimal - the logic lives in the repository.
- The first failure demonstrated how console output plus stage granularity localize defects quickly.
- Jenkins attributing the build to 'meenakshi' and the commit to 'meenakshi16rp' confirms the identity configuration end to end.